

Marketing ROI & Budget Reallocation Analysis

Capstone Project Report

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1. Introduction:

Marketing teams invest significant budgets in multiple campaigns across different channels. However, identifying which campaigns generate the highest return on investment (ROI) can be challenging without proper data analysis.

This project aims to analyze marketing campaign performance and measure ROI using a Python-based data pipeline and analytics framework. The system processes raw marketing data and generates insights that help optimize marketing budget allocation.

2. Project Objective:

The main objectives of this project are:

- To build a data pipeline for processing marketing data.
 - To analyze campaign performance across marketing channels.
 - To compute Return on Investment (ROI) for marketing campaigns.
 - To identify high-performing and underperforming campaigns.
 - To provide insights for marketing budget optimization.
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3. Dataset Description:

The project uses simulated marketing datasets consisting of multiple CSV files.

Dataset	Description
sessions.csv	Website sessions including user information, device type, marketing channel, and campaign ID
orders.csv	Customer purchase orders generated from sessions
order_items.csv	Product details including price and quantity
campaigns.csv	Campaign metadata
ad_spend_daily.csv	Daily advertising spends per campaign

These datasets together allow analysis of the relationship between marketing spend and revenue generation.

4. Data Pipeline Architecture:

The system follows a structured data processing pipeline:

Raw CSV Data

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ETL Pipeline (Python)

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Fact Tables

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ROI Analysis

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Visualization & Insights

The pipeline ensures that raw data is transformed into clean analytical datasets suitable for performance evaluation.

5. ETL Process:

The ETL (Extract, Transform, Load) pipeline performs the following steps:

Extract

Raw datasets are loaded using the Python **Pandas** library.

Transform

The transformation process includes:

- Merging session data with order and product data
- Calculating revenue for each order
- Creating structured fact tables for campaign and channel analysis

- Aggregating revenue and order counts by date

Load

The processed datasets are exported as analytical tables:

- **fact_sessions.csv**
- **fact_campaign_daily.csv**
- **fact_channel_daily.csv**

These tables serve as the foundation for ROI analysis.

6. ROI Analysis:

Return on Investment (ROI) is calculated to evaluate marketing effectiveness.

ROI Formula

ROI = Revenue / Advertising Spend

Campaign-level ROI results are stored in:

campaign_roi_summary.csv

This allows identification of the most profitable campaigns.

7. Data Visualization:

A campaign performance visualization was generated using **Matplotlib**.

Output file:

campaign_roi_chart.png

The chart compares ROI across different campaigns and highlights:

- High-performing campaigns
 - Underperforming campaigns
 - Opportunities for marketing budget reallocation
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8. Key Insights:

Based on the analysis, several important insights were identified:

High ROI Campaigns:

Some campaigns generate very high ROI (10x–16x), indicating strong marketing efficiency.

Moderate ROI Campaigns:

Several campaigns produce moderate ROI (4x–6x), suggesting stable performance with room for optimization.

Low ROI Campaigns:

Certain campaigns produce ROI below 2x, indicating inefficient marketing spend.

Budget Reallocation Opportunity:

Marketing budgets should be shifted from low-performing campaigns to high-performing campaigns to maximize revenue.

9. Technology Stack:

The project uses the following tools and technologies:

Tool	Purpose
Python	Data processing and analysis
Pandas	Data transformation and aggregation
Matplotlib	Data visualization
VS Code	Development environment

10. Project Folder Structure:

Marketing ROI Capstone

```
data/
  sessions.csv
  orders.csv
  order_items.csv
  campaigns.csv
  ad_spend_daily.csv

etl/
  etl_pipeline.py

analysis/
  fact_sessions.csv
```

fact_campaign_daily.csv
fact_channel_daily.csv
campaign_roi_summary.csv
campaign_roi_chart.png
roi_analysis.py
visualize_roi.py

11. Conclusion:

This project demonstrates how data engineering and analytics techniques can transform raw marketing data into actionable business insights.

By building an ETL pipeline and performing ROI analysis, organizations can identify high-performing campaigns and reallocate marketing budgets more effectively to maximize return on investment.